Today

- Delta function potential
- Reflection and transmission

2.4 Delta function potential

Consider a particle of mass m in a delta function potential well,

$$V(x) = -\alpha\delta(x), \tag{2.4.1}$$

where α is a positive constant. A schematic figure of the potential is shown below.

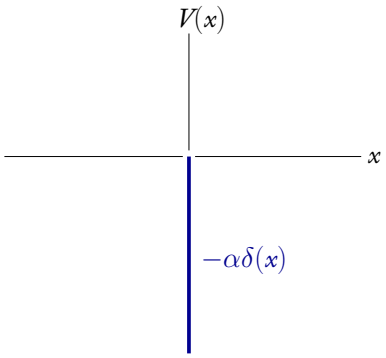


Figure 2.4.1: Delta function potential well.

To construct any bound state wavefunctions and the corresponding energies, we solve the Schrödinger equation on the left and the right sides of $x = 0$ and impose the continuity of $\psi(x)$ and the discontinuity of $\psi'(x)$ across it. Note that now that the potential has infinite discontinuity at the origin, $\psi'(x)$ is not continuous. Denoting the region $x < 0$ by I and the region $x > 0$ by II, the Schrödinger equation in region I is

$$\psi_I''(x) - \kappa^2\psi_I(x) = 0, \qquad \kappa \equiv \sqrt{-\frac{2mE}{\hbar^2}}, \tag{2.4.2}$$

where $E < 0$ for a bound state. The general solution is $\psi_I(x) = Ae^{-\kappa x} + Be^{\kappa x}$. Because the wavefunction has to vanish at $x = -\infty$, in order for it to be normalizable, we set $A = 0$, and we have

$$\psi_I(x) = Be^{\kappa x}. \tag{2.4.3}$$

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The Schrödinger equation and the general solution in region *II* are exactly the same as in region *I*. The only difference is that the wavefunction in region *II* must vanish at $x = \infty$, which sets $B = 0$, and we have

$$\psi_{II}(x) = Ae^{-\kappa x}. \tag{2.4.4}$$

Continuity of the wavefunction across $x = 0$ imposes

$$A = B. \tag{2.4.5}$$

The discontinuity of $\psi'(x)$ across $x = 0$ follows from

$$\begin{aligned} \psi'_{II}(0) - \psi'_{I}(0) &= \lim_{\epsilon\to 0}\int_{-\epsilon}^{\epsilon}\psi''(x)dx = \lim_{\epsilon\to 0}\int_{-\epsilon}^{\epsilon}\frac{2m}{\hbar^2}(V(x)-E)\psi(x)dx \\ &= -\lim_{\epsilon\to 0}\frac{2m}{\hbar^2}\int_{-\epsilon}^{+\epsilon}\alpha\delta(x)\psi(x)dx = -\frac{2m}{\hbar^2}\alpha\psi(0) = -\frac{2m}{\hbar^2}\alpha A, \end{aligned} \tag{2.4.6}$$

On the other hand, taking the derivatives directly, $\psi'_{II}(0) - \psi'_{I}(0) = -\kappa A - \kappa B$. Therefore,

$$\kappa A + \kappa B = \frac{2m}{\hbar^2}\alpha A. \tag{2.4.7}$$

Solving (2.4.5) and (2.4.7), we have

$$\kappa = \frac{m\alpha}{\hbar^2}, \tag{2.4.8}$$

which using the definition of κ gives

$$E = -\frac{m\alpha^2}{2\hbar^2}. \tag{2.4.9}$$

Because there is only one solution, the delta function potential well supports one and only one bound state. Because $A = B$, the wavefunction decays symmetrically and exponentially on both sides. Writing $\psi_I(x) = Ae^{\kappa x}$ and $\psi_{II}(x) = Ae^{-\kappa x}$, and normalizing, $1 = \int_{-\infty}^{0} A^2 e^{2\kappa x} + \int_{0}^{\infty} A^2 e^{-2\kappa x} = A^2\frac{1}{2\kappa} + A^2\frac{1}{2\kappa} = \frac{A^2}{\kappa}$, which gives $A = \sqrt{\kappa}$. Therefore, the normalized wavefunction is

$$\psi(x) = \begin{cases} \sqrt{\kappa}e^{-\kappa x} = \sqrt{\frac{m\alpha}{\hbar^2}}e^{-\frac{m\alpha}{\hbar^2}x}, & x > 0 \\ \sqrt{\kappa}e^{\kappa x} = \sqrt{\frac{m\alpha}{\hbar^2}}e^{\frac{m\alpha}{\hbar^2}x}, & x < 0. \end{cases}$$

The energy level and the wavefunction are shown in the schematic figure below.

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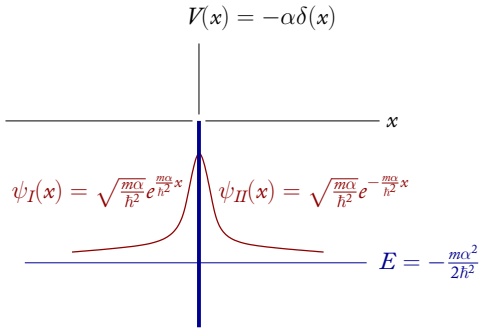


Figure 2.4.2: Energy and wavefunction of the delta function potential well.

Question 2.4.1: Can we have an example of a system that can be represented by an attractive delta function potential?

Answer: An exact delta function potential is a simplified mathematical idealization. However there are physical systems that can be approximated by it. One such system, perhaps the best example, is what is called Feshbach resonance in ultracold gases composed of alkali atoms such as rubidium, sodium, lithium, or cesium. When cooled to nanokelvin range while the number density of atoms is around 10^{11} to 10^{15} per cm^3, then the de Broglie wavelength of the atoms becomes much larger than the range of intermolecular interaction. This turns out to lead to and effective zero-range attractive delta function potential, $V(x) = -\alpha\delta(x)$ in one dimension, to be a good approximation. The strength of the potential α can be tuned by varying an applied magnetic field to produce a deep resonant potential well resulting in a bound state of two atoms called a dimer molecule.

Another example is the periodic potential from atomic ions in conductors, which can be crudely approximated by a periodic attractive delta function potential (called Dirac comb potential) to gain a qualitative picture of conductivity and band structures in a simplified mathematical setting.

2.5 Reflection and transmission

Now we want to consider the case of a particle of energy $E > 0$ approaching the well from the left of the finite square well potential whose bound states we have investigated. Now the particle has positive kinetic energy everywhere above the potential and it moves along from $x = -\infty$ to $x = \infty$.

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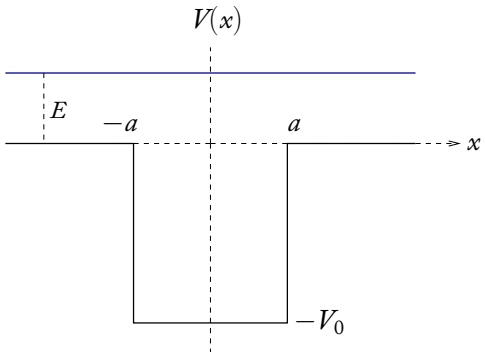


Figure 2.5.1: A particle with energy above an infinite square well.

Question 2.5.1: For a particle represented by a plane wave with energy $E > 0$ approaching the well from the left of an finite square well, (A). The wavefunction vanishes at $x = \pm\infty$. (B). The particle is partly reflected and partly transmitted at $x = -a$. (C). The particle is partly reflected and partly transmitted at $x = a$. (D). The particle is partly reflected and partly transmitted at all points it passes through. (E). The speed of the particle increases when it is between $x = -a$ and $x = a$. (F). The wavelength of the particle decreases when it is between $x = -a$ and $x = a$. (G). The particle moves along without any reflection anywhere.

Answer: All except (A) and (D) are answers. Because the particle is free and unbounded represented by a plane wave, it can propagate along from $x = -\infty$ to $x = \infty$. Because the potential changes at $x = \pm a$, the wave (and therefore the particle) is is partly reflected and partly transmitted $x = \pm a$. Note that an incoming plane wave Ae^{ikx} cannot match to a continuing plane wave $Ae^{i\kappa x}$ at $x = -a$ with the same amplitude A while $k \neq \kappa$. The boundary conditions can lead to consistent equations only with considerations of reflection and transmission. There is no reflection at other points where the potential does not change. The kinetic energy of the particle (and its speed) increases, and its wavelength decreases when it is above the well, recall that $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m(E-V)}}$.

Question 2.5.2: The coefficient of transmission T should in general be defined as (A). $\frac{J_{tra}}{J_{inc}}$, (B). $\left|\frac{\psi_{tra}}{\psi_{inc}}\right|^2$, (C). Either one of the above.

Answer: The answer is (A). $\frac{J_{tra}}{J_{inc}}$. The amplitude tells us only about the density of particles (number per unit length, $\mathcal{P} = \frac{dN}{dx}$). We need the number of particles per unit time flowing through a point in order to compare number of particles transmitted to that of the incident, $\frac{dN}{dt} = \frac{dN}{dx}\frac{dx}{dt} = \mathcal{P}v = |\psi|^2 v = J$. Thus, if the

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speeds are different, the ratio of the amplitudes cannot capture the ratio of rate of particle flows. In other words, $\frac{\partial \mathcal{P}}{\partial t} = \frac{\partial \mathcal{J}}{\partial x}$, and integrating both sides between two points a and b on x, $\frac{dP}{dt} = -(J(b) - J(a))$. Therefore, the rate of particle flow through a point is given by the difference in the probability currents between the two points. Considering a wave of the form $\psi(x) = Ae^{ikx}$, and using the definition of J, we have $J = \frac{\hbar k}{m}|A|^2 = \frac{p}{m}|A|^2 = v|A|^2$, and the velocity matters, since a smaller probability density could result in larger flux of particles if the speed is large. When the speeds are the same on both sides, the velocity cancels out and the transmission coefficient $\frac{J_{tra}}{J_{inc}}$ reduces to $\left|\frac{A_{tra}}{A_{inc}}\right|^2$.

Recalling that the probability current describes the particle current, the rate of particle flow though a given point, we should note that in order to compute the probability for reflection or transmission, we compare the corresponding currents. The coefficients of reflection R and transmission T are defined as

$$R = \frac{J_{ref}}{J_{inc}}, \qquad T = \frac{J_{tra}}{J_{inc}}, \tag{2.5.1}$$

where J_{inc} is the probability current for the incident wave, J_{ref} is the probability current for the reflected wave, and J_{tra} is the probability current for the transmitted wave. Comparing the probability densities only does not provide a correct answer, unless the speed of the particle is the same in the corresponding regions.

Considering the finite potential well shown in the figure above, and denoting the region in $x < -a$ by I, that within $-a \leq x \leq a$ by II, and the region in $x > a$ by III, the solutions to the Schrödinger equation in each region can be written down immediately,

$$\psi_I(x) = Ae^{ikx} + Be^{-ikx}, \tag{2.5.2}$$

$$\psi_{II}(x) = C\sin\kappa x + D\cos\kappa x, \tag{2.5.3}$$

$$\psi_{III}(x) = Fe^{ikx}, \tag{2.5.4}$$

where

$$\kappa \equiv \sqrt{\frac{2mE}{\hbar^2}}, \qquad \kappa \equiv \sqrt{\frac{2m(E+V_0)}{\hbar^2}}, \tag{2.5.5}$$

A is the amplitude of the incident wave, B is the amplitude of the reflected wave, and F is the amplitude of the transmitted wave. There is no reflected wave that propagates to the left in region III. We have chosen sin and cos in region II, because we will find that we have perfect transmission when the energy of the incident wave corresponds to a bound state of the corresponding infinite square well and the expressions become simpler. Applying the continuity

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of $\psi(x)$ and $\psi'(x)$ at $x = \pm a$ gives the following four equations,

$$\begin{aligned} Ae^{-ika} + Be^{ika} &= -C\sin\kappa a + D\cos\kappa a, && (2.5.6)\\ ik\left(Ae^{-ika} - Be^{ika}\right) &= \kappa\left(C\cos\kappa a + D\sin\kappa a\right), && (2.5.7)\\ C\sin\kappa a + D\cos\kappa a &= Fe^{ika}, && (2.5.8)\\ \kappa\left(C\cos\kappa a - D\sin\kappa a\right) &= ikFe^{ika} && (2.5.9) \end{aligned}$$

We have four equation which we can solve for four variables B, C, D, and F in terms of the amplitude A of the incident wave. Our interest is in the amplitudes B and F for the reflected and the transmitted waves, and we write the solutions for B and F,

$$F = \frac{e^{-2ika}}{\cos(2\kappa a) - i\frac{k^2+\kappa^2}{2k\kappa}\sin(2\kappa a)}A \tag{2.5.10}$$

and

$$B = i\frac{\sin(2\kappa a)}{2k\kappa}(\kappa^2 - k^2)F. \tag{2.5.11}$$

The transmission coefficient is obtained using Ae^{ikx} for the incident wave and Fe^{ikx} for the transmitted wave, which gives

$$T = \frac{J_{tra}}{J_{inc}} = \left|\frac{F}{A}\right|^2 = \frac{1}{1 + \frac{V_0^2}{4E(E+V_0)}\sin^2\left(\frac{2a}{\hbar}\sqrt{2m(E+V_0)}\right)}, \tag{2.5.12}$$

and the transmission coefficient oscillates with the energy of the incident particle. Moreover,

$$R = 1 - T. \tag{2.5.13}$$

The transmission coefficient oscillates with energy as shown in the figure below.

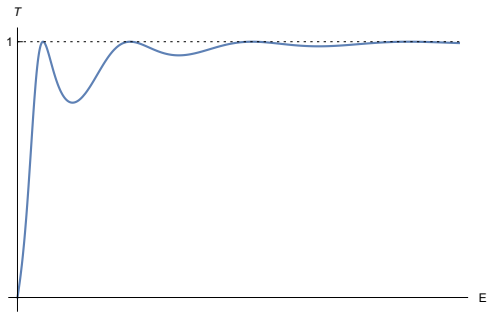


Figure 2.5.2: Transmission coefficient for a particle over a square potential well.

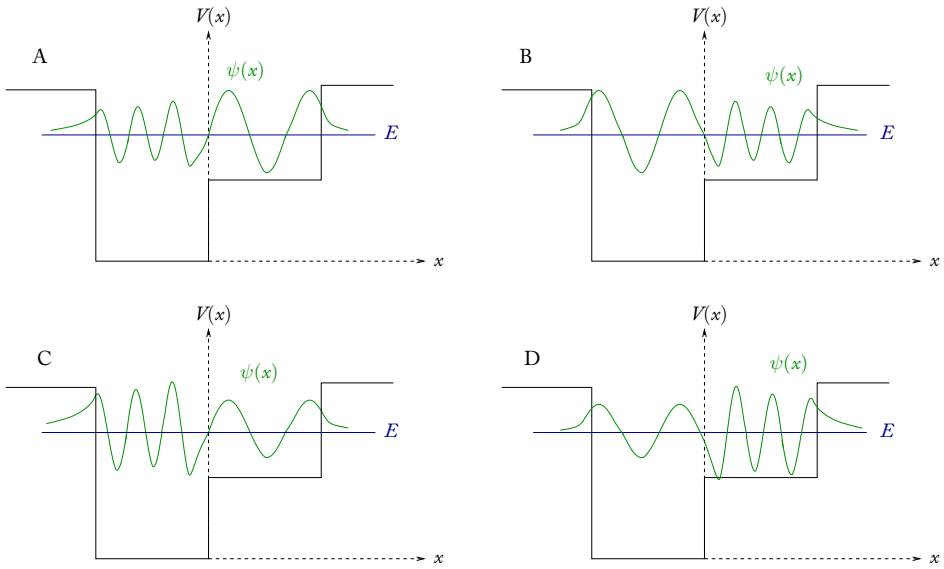
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Question 2.5.3: Discuss why the transmission coefficient oscillates with the energy of the incident particle and what the peaks correspond to.

Answer: The reflection and transmission depend on the wavelength of the particle (which is related to its energy). It turns out a smaller energy that corresponds to an appropriate wavelength could produce a higher coefficient of transmission than a higher energy. In particular, we see that we have a resonant perfect transmission without reflection if the energy is such that $\sin\left(\frac{2a}{\hbar}\sqrt{2m(E+V_0)}\right) = 0$, which occurs when $\frac{2a}{\hbar}\sqrt{2m(E+V_0)} = n\pi$, $n = 1, 2, 3, \ldots$ or $E_n = -V_0 + \frac{\hbar^2\pi^2n^2}{8ma^2}$. These are exactly the bound states of a corresponding infinite square well. In other words, when the kinetic energy $\frac{\hbar^2\pi^2n^2}{8ma^2}$ in the total energy $E = -V_0 + \frac{\hbar^2\pi^2n^2}{8ma^2}$ of the incident particle when above the well, corresponds to one of the bound states of the infinite well of the same width (or when width of the well is a multiple of the de Broglie wavelength of the particle) we have perfect transmission without reflection. This effect is called the Ramsauer–Townsend effect. Slow electrons fired through inert gases such as *Ar*, *Kr*, or *Xe*, were observed to undergo perfect transmission for specific values of energy, as shown in the figure above. At these energies, the reflection from $x = \pm a$ match and produce standing stationary modes above the well. Note that these correspond to energies of a corresponding infinite square well and not of the finite well, since the particle has positive energy above the well and not trapped in the well, the wavelength just needs to match the well so that the reflections from the two ends balance and create a standing mode in between.

Question 2.5.4: Which one of the following best describes the wavefunction of a bound particle of energy E in the potential well $V(x)$?

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Answer: The answer is (A). The particle has higher kinetic energy, higher momentum, and shorter wavelength in the left region where the well is deeper. Because the particle is faster, the probability amplitude for finding it there is also smaller, resulting in smaller amplitude, in comparison to the right side of the well, where both the wavelength and the amplitude are larger.

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